

Website build guide

From setup to launch

An implementation brief for Claude Code or Codex
Revised seven-tab scope | 2 October 2026

Read this first

Build Nigeria's Beautiful Minds (NBM): a Nigeria-first platform connecting researchers, innovators, students and Nigerians abroad through public work, conversation, mentorship and sponsorship. Make it credible, welcoming and socially engaging, with a clear route from discovering an idea to developing a supported project.

This brief supersedes the earlier five-section navigation. The main tabs, in order, are Home, About, Get Involved, Exhibitions, Discussion, Mentorship and Sponsorship. Home is a distinct page. About has the four specified dropdown items. Get Involved has a landing page and four participation routes.

Instruction to the coding agent

Treat this PDF as the product specification. First inspect the supplied repository and existing deployment. Preserve working functionality and content; reuse sound code. Implement the phases below in dependency order and verify each against its completion gate. A visual mockup, static demonstration or simulated success message is not a completed feature.

Maintain a build checklist recording complete, in progress, blocked and not started items. At each phase report the changed files, migrations, evidence from checks, remaining blockers and next step. Continue with authorized reversible work without requesting approval after every phase. Ask focused questions only where a missing business decision or access prevents safe progress.

Never invent credentials, successful payments, real users, partnerships, team biographies, grants or impact statistics. Seed records belong in development or staging and must be labelled there. If a provider is unavailable, finish the integration and testable local work, label the live path blocked, and identify the exact configuration needed.

Scope boundary

The first release includes photo/video exhibitions, moderated comments and reactions, text discussions, live audio rooms, human-assisted mentorship matching, general contributions, project sponsorship and staff administration. Native mobile apps, algorithmic recommendations, automatic matching, unrestricted direct messages and audio recordings are deferred.

This is a new developer handoff document. The prior concept remains background; where it conflicts with this brief, use the seven-tab scope here. The name NBM remains the working brand unless the founder changes it.

Site map and navigation contract

Main tab	Required destinations
Home	/
About (dropdown)	About us: /about News: /about/news Our team: /about/team Privacy Policy: /about/privacy
Get Involved	Landing: /get-involved Contribute: /get-involved/contribute Volunteer: /get-involved/volunteer Join our mailing list: /get-involved/mailling-list Contact us: /get-involved/contact
Exhibitions	/exhibitions; project detail /exhibitions/[slug]; signed-in create/edit screens
Discussion	/discussion; text threads /discussion/[id]; audio rooms /discussion/rooms/[id]
Mentorship	/mentorship; opportunities /mentorship/opportunities/[id]; private applications and matches
Sponsorship	/sponsorship; approved campaigns /sponsorship/[slug]; support requests

Render About as an accessible dropdown with exactly the four entries above. Get Involved must be clickable and may also expose its four links in a dropdown. On mobile, use a collapsible menu with expandable submenu buttons; do not depend on hover. Sign in, Join and the signed-in account menu are utility controls, not extra main tabs.

Supporting routes. Provide profile, account settings, notifications, saved items and a private workspace under the account menu. Staff tools live under /admin. Terms, community standards and reporting links may sit in the footer; they do not add main tabs.

One contribution system. Contribute supports the general program. Sponsorship supports a specified project, cohort or placement and accepts institutional or in-kind enquiries. Both use the same contribution records and payment integration, distinguished by the funding destination. Do not create two independent donation systems.

One content system. News is editorial content published by staff. Exhibitions are member project posts. A news article can link to an exhibition, but the same content should not be copied into disconnected databases.

Use the supplied example pages only as navigation and content-structure inspiration. Write original NBM copy, use authorized assets, and do not reuse another organization's identity, team details, achievements, contact information or privacy policy. The public site must not name the reference organization.

Phase 0 | Audit and establish the work area

Goal: Understand what exists before changing it. Required input: repository access or an empty project directory and the intended deployment context.

0.1 Inspect. Read repository instructions, package manifests, routes, authentication, database schema, migrations, storage integrations, audio integrations, tests and deployment configuration. Record which features work, which are partial and which are absent. Inspect secret names only; do not print secret values.

0.2 Protect. Create a working branch or isolated checkout. Preserve unrelated edits. Take a database backup before any production migration. Never delete or recreate the live database to make a migration succeed.

0.3 Document. Create docs/build-checklist.md, docs/architecture.md and docs/decisions.md. Map the current code to this specification. Record the existing provider accounts, environments and unresolved founder inputs.

Completion gate: The project starts locally, or its specific startup blocker is documented; baseline checks are recorded; the agent can explain what will be reused and what needs building.

Phase 1 | Confirm architecture and dependencies

1.1 Keep the stack practical. Use the existing supported framework where viable. Separate the responsive interface, server-side business logic, managed authentication, relational database, object storage, live-audio service, email delivery and hosted payments. Reuse existing service accounts where authorized.

1.2 Verify integrations. For each selected service, consult its current official documentation, supported SDK and pricing before implementation. Confirm the relevant region, account eligibility, spending limits and development environment. Do not assume a free tier covers the pilot.

1.3 Set environments. Create local, staging and production configurations. Commit an .env.example containing names and descriptions only. Configure server-only credentials, allowed origins, authentication callbacks and webhook URLs separately for each environment.

1.4 Plan operations. Document expected media volume, audio participant-minutes, email volume and payment processing costs. Use adjustable quotas and alerts. Initial development effort is separate from recurring service costs and research grant funds.

Completion gate: An architecture map, environment checklist and reproducible local setup exist. Available services pass connection checks; missing credentials are named blockers with no fake successful state.

Founder inputs to track: Domain and hosting access; final brand assets; factual team details; contact recipient; approved policies; payment account and funding rules; mentor reviewers; initial project content. Missing copy may remain clearly marked in staging, but not in a public launch.

Phase 2 | Data model, accounts and permissions

Data group	Minimum records and constraints
Identity	Accounts, public profiles, roles, consent records and notification settings. Keep email and private contact details out of public profiles.
Content	Pages, news, team members, exhibitions, media, comments, reactions and saves. Link author and status; index published dates and topics.
Conversation	Threads, messages, audio rooms, membership, speaker requests and reports. Bind room ownership to authenticated identities.
Mentorship	Mentor applications, verified mentor profiles, opportunities, student applications, matches and milestones. Restrict private documents.
Support	Campaigns, contribution intents, provider transactions/events, refunds, grant disbursements and milestone reports. Use unique provider IDs and explicit currencies.
Operations	Volunteer applications, mailing consent, contact requests, moderation actions and audit logs. Define retention and deletion behavior.

2.1 Create migrations. Use versioned migrations, foreign keys, required fields, indexes and uniqueness constraints. Enforce one active reaction per member per exhibition and one payment event per provider/event ID. Use integer minor currency units for money. Store timestamps consistently and display local time zones.

2.2 Build authentication. Implement registration, email verification, sign-in, sign-out and recovery using the chosen auth service. Validate redirects. Add rate limits and stronger authentication for privileged accounts. Members may edit only their own profiles.

2.3 Enforce access. Visitors read approved public content. Members interact and submit their own work. Mentors see only applications assigned or explicitly shared with them. Coordinators manage matches; moderators manage reports; finance staff manage funding records; administrators assign privileged roles. Staff roles may be combined for a small pilot but permissions remain explicit.

Completion gate: Test two unrelated accounts, a reviewer and an unauthenticated visitor. Direct requests cannot read another applicant's documents, grant staff privileges, publish pending work or alter another user's records. Permissions must be enforced server-side and at the data layer where supported.

Phase 3 | Shared interface and reusable states

Create the header, mobile navigation, dropdowns, footer, buttons, forms, cards, dialogs, pagination and notifications. Use a restrained green-and-neutral palette, readable type and authentic project imagery. Build loading, empty, error, success and disabled states. Forms preserve inputs after recoverable errors.

Completion gate: All seven navigation items work with keyboard, touch and mouse. Layouts remain usable at narrow mobile widths. Focus, labels, contrast and errors are clear; no horizontal page overflow or decorative elements obscure content.

Phase 4 | Home and About

4.1 Home. Build a concise hero introducing NBM and connecting Nigerians at home and abroad. Primary actions: Explore projects and Join the community. Follow with featured exhibitions, upcoming/live discussions, mentorship opportunities, sponsorship opportunities, recent news and a Get Involved invitation.

Populate Home from published records, not hard-coded duplicate cards. Use a staff-selected featured exhibition when available, otherwise the newest approved exhibition. Hide unavailable modules gracefully. Do not display fabricated membership counts, donors, testimonials or impact metrics. The logo returns to Home.

4.2 About us. Explain mission, vision, Nigeria-first scope, future African expansion, who can join and how exhibition, discussion, mentorship and sponsorship connect. State the relationship with Optimais Labs only as confirmed by the founder.

4.3 News. Create a searchable or category-filtered article list and detail pages with title, excerpt, author, date, body and optional image. Staff can draft, preview, publish and archive. Drafts are never public or indexed.

4.4 Our team. Show approved names, roles, photographs and short biographies. Group staff, advisers and organizing committees when real members are supplied. Provide an honest empty state in staging; do not invent people or affiliations.

4.5 Privacy Policy. Implement a versioned, editable policy page with effective date and contact route. Its content must describe the actual data collected, purposes, service providers, audio behavior, retention and requests process. Have the founder obtain appropriate review before publication. Do not copy the reference policy or assume it matches NBM.

Completion gate: Editors can update pages, news and team records without code changes. Published content appears on the right public page and Home modules update correctly. Preview access does not expose drafts to visitors.

Phase 5 | Get Involved

Create a landing page beginning: “There are many ways to support research, innovation and the next generation of Nigerian talent.” Present four clear cards: Contribute, Volunteer, Join our mailing list and Contact us.

Contribute. Explain what general contributions support and link to the shared hosted payment flow built in Phase 10. Include one-time support initially. Add recurring contributions only if configured and tested with cancellation handling. Keep live payments disabled until the finance launch gate passes.

Volunteer. Collect name, email, interests, committee preference, relevant skills and availability. Offer a small editable list of committees such as events, mentorship, outreach and project review. Save applications privately for staff review and communicate receipt, not automatic acceptance.

Dependency: Build the remaining forms and delivery behavior in Phase 5 continued, before enabling the public Get Involved routes.

Phase 5 continued | Participation forms

Mailing list. Collect email and optional name with clear opt-in wording. Use a confirmation link before activating subscriptions. Store consent time, source and policy version. Handle existing subscribers without duplicates and provide a working unsubscribe link. Joining the platform does not automatically subscribe a member.

Contact us. Collect name, email, subject and message. Save or reliably queue the submission before displaying success. Notify the configured recipient through the email service. Show accurate failure states and retry delivery safely without generating duplicate requests.

Staff handling. Provide private lists for volunteer and contact submissions, with status, assignee and notes. Volunteer statuses: submitted, reviewing, accepted, declined or withdrawn. Export only for authorized staff. Do not expose applicants' personal details publicly.

Abuse protection. Validate on the server, sanitize display, rate-limit submissions and use appropriate bot protection. Keep email credentials server-side. Log delivery status without recording unnecessary message contents or sensitive fields.

Completion gate: Submissions persist; authorized staff can review them; intended notifications arrive in a test inbox; opt-in and unsubscribe work; duplicate requests do not duplicate subscriptions. A delivery failure is distinguishable from a successfully saved form. Do not send test messages to real applicants or volunteers.

Phase 6 | Exhibitions: media and publication

6.1 Build the author flow. Signed-in members create drafts with title, description, topic, project stage, team credits, optional project link and photos or a short video. Include alt text or a useful image description, upload progress, retry and a preview. Allow editing, submission and withdrawal.

6.2 Protect uploads. Validate file signature, type and size on the server. Use scoped upload permissions, unguessable storage keys, malware checks and controlled publication. Keep pending media private; do not rely on hidden links. Delete abandoned uploads through a scheduled cleanup process.

6.3 Publish fairly. Member submissions enter pending review. Editors and administrators may publish directly. Staff approve, reject with a reason or remove published content. Material edits to an approved member post require renewed review; keep the last approved version visible until replacement approval.

Proposed pilot limits: Up to five images per post at 10 MB each, or one video up to 100 MB and three minutes. Make limits configurable and enforce actual size/duration; verify browser playback before accepting formats. No video autoplay. These are product defaults to validate during the beta.

Completion gate: A member can submit real media, a moderator can approve it, and visitors see only the approved version. Invalid files, oversize uploads and direct access to pending assets are rejected. Failed uploads can be retried without duplicate posts.

Phase 6 continued | Exhibition engagement

Build the feed and detail page with author, date, topic, project stage, media and description. Support latest-first ordering, topic filters and pagination. Add comments, reply threads, saved posts, share links and one selected emoji reaction per person. Changing or removing a reaction updates counts accurately.

Authors may link their exhibition to an existing discussion, mentorship opportunity or approved sponsorship campaign. Members can report content; staff can review reports and record reasons for action. Sanitized user text must not execute scripts. Removed or rejected posts must disappear from public lists and search results.

Completion gate: Two members can comment and react without duplicate counts; permissions prevent editing another author's post; spam limits work; moderation removal also removes public media access as designed. Home displays only approved project content.

Phase 7 | Discussion: text conversations

7.1 Create and discover. Members create a conversation with title, topic and opening message. Provide recent, active and topic filters. Each thread has a stable address, host identity and clear participation controls. News articles, exhibitions and research links can be referenced without copying private content.

7.2 Participate. Signed-in users add comments or replies, edit their own messages and see new activity through realtime updates or efficient polling. Paginate older messages. Prevent duplicate sends after retries; retain an unsent draft when the connection drops.

7.3 Moderate. A host can close or lock their own conversation and report/remove disruptive messages within defined permissions. Members can report and block abusive accounts; document the limits of blocking in public rooms. Staff can intervene, archive and audit actions. A host cannot control another host's thread.

7.4 Connect. Allow a text conversation to reference a scheduled audio room and retain a written summary afterward. Notify participants about replies according to their settings. Do not create unrestricted private messaging for the initial release.

Completion gate: Two browsers exchange messages reliably; retries do not duplicate records; lost connections recover; locked threads reject new messages server-side. Visitors can read approved public threads but must sign in to participate.

Social quality: Invite project-specific questions, findings, lessons and collaboration requests. Avoid making academic seniority a prerequisite for contributing. Keep the host controls visible and make report actions reachable on mobile.

Phase 8 | Discussion: live audio rooms

Dependency: Authentication, thread permissions and the selected audio service must work first. Integrate live audio inside Discussion; an external meeting link alone does not satisfy this requirement.

8.1 Room lifecycle. Members schedule or start a room with title, description, topic and time zone-aware start time. Use scheduled, live, ended and cancelled states. Show host, participant count, live status and a linked text channel. Create service rooms and tokens from the server, not the browser.

8.2 Listener-first entry. A signed-in member joins as a listener with audio publishing disabled by service permissions. Listening should not request microphone access. Request the microphone only when the person becomes a speaker and chooses to enable it. Provide clear join, leave, mute and device-selection controls.

8.3 Host control. Implement request-to-speak, approve/decline, demote to listener, mute, remove and end-room actions. Changes must update service-side permissions immediately, not just UI buttons. Removed members cannot simply reuse a token to rejoin while blocked. Hosts must never remotely unmute a person's microphone.

8.4 Access and recovery. Issue short-lived room-scoped tokens after membership, room-state and role checks. A user cannot mint host permissions or enter an ended/private room through a crafted request. Handle denied microphone access, missing devices, weak network and reconnect without duplicate identities.

8.5 Cost and lifecycle. Use a configurable initial cap of 25 participants and 5 simultaneous speakers. Define a host-disconnect grace period with visible countdown; end an abandoned room after it expires. Stop idle rooms and reconcile room states after service failures. Track participant-minutes and configure spending alerts.

8.6 Privacy. Do not record audio in the first release. Explain this and any retained participation metadata. Provide a text alternative for room participation. Any later recording feature requires explicit design, consent, retention and access decisions.

Completion gate: Test at least three real browser sessions as host, listener and speaker. Verify listener publishing is denied even through modified client requests. Test promotion, removal, reconnect, host loss, end-room and capacity handling. Validate the chosen capacity with a staging load check before advertising it.

Evidence: Record supported browser/device results and known limitations, including mobile microphone permission and background-tab behavior. Do not mark audio complete solely because a room can be created or its interface renders.

Phase 9 | Mentorship and research matching

9.1 Mentor onboarding. Collect expertise, affiliation, project interests, availability, location, working mode and capacity. Staff verify the relevant role before displaying a verified label. Distinguish a research PI from a technical mentor. Do not promise institutional supervision without the institution's agreement.

9.2 Opportunities. Mentors or staff create research/innovation opportunities with project scope, skills, learning goals, expected duration, hours, location, facilities and funding availability. Publish only approved opportunities and close those with no capacity.

9.3 Student applications. Collect interests, experience, motivation, availability and a CV or portfolio where needed. Explain who will see the application. Permit draft, submit and withdraw; preserve a dated submitted version. Restrict document URLs and downloads to authorized reviewers.

9.4 Human-assisted matching. A coordinator filters by subject, skills, goals, location and capacity, then proposes a match. Both sides must accept. One acceptance alone must not activate the placement. Prevent overbooking when offers are accepted concurrently.

9.5 Project workspace. After acceptance, capture the agreed brief, supervisor, resources, meeting cadence, milestones and expected output. Provide private progress updates, completion and rematch requests. Only a separately approved public summary may become an exhibition.

State	Required behavior
Submitted / reviewing	Applicant and authorized reviewers see the application; applicant can withdraw.
Proposed match	Coordinator records the proposal and both parties receive a private action.
Accepted / active	Both parties agree; capacity is reserved and the project brief becomes active.
Declined / expired	Record outcome and release any reserved capacity; coordinator can propose another match.
Completed / closed	Record deliverables and feedback; keep private documents protected.
Rematch requested	Coordinator reviews the issue and manages closure or reassignment.

Completion gate: Walk one student and one mentor through application, review, mutual acceptance, milestones and completion. Verify a second mentor cannot access the application, a withdrawn application cannot be activated, and capacity cannot be exceeded by parallel accepts.

Honest claims: Matching does not guarantee admission, employment, funding or a publication. Show the actual funding status and agreed responsibilities for each opportunity. Keep monetary awards in Sponsorship records rather than duplicating them here.

Phase 10 | Sponsorship and shared contributions

10.1 Campaign review. Build support-request intake and staff review. Approved campaigns show purpose, team, intended beneficiaries, budget, currency, milestones, funding target, deadline and update history. Staff define what happens when a target is missed or exceeded before opening contributions.

10.2 Shared payment flow. General Contribute creates a contribution to the general fund. Sponsorship creates one tied to a campaign ID. Validate destination, amount and currency on the server and create a provider checkout session. A return URL displays pending until the server confirms the transaction.

10.3 Trusted confirmation. Verify provider webhook signatures and transaction details. Deduplicate by event ID and transaction ID; process retries and out-of-order events safely. Do not trust client-supplied success or totals. Maintain initiated, pending, succeeded, failed, refunded and disputed states as supported.

10.4 Accurate accounting. Record amount, currency, fees and provider references. Keep pledged, received and disbursed values separate. Display totals in their actual currency; never add different currencies without an explicit documented conversion method. Refunds and reversals must adjust the relevant public totals.

10.5 Donor experience. Allow hosted checkout without a community account if the provider supports it. Collect only required donor details. Public acknowledgement is opt-in; anonymous public display does not erase internal payment records. Send a payment acknowledgement, without claiming tax deductibility or issuing unsupported tax certificates.

10.6 Institutional support. Provide forms for company/government partnerships and in-kind support. Route these to staff review; submitting a pledge or equipment offer must not increase confirmed cash totals. Grant releases require finance review and milestone evidence, not automatic transfer after a user uploads a report.

Completion gate: In a payment sandbox, test success, failure, cancellation, duplicate webhook, delayed confirmation, refund and currency mismatch. Confirm one transaction produces one contribution and one acknowledgement. Verify donor records are private and only authorized finance staff can record disbursements.

Live activation: Before accepting real money, the founder confirms the receiving organization/account, provider eligibility, currencies, refund rules and finance owner. Use a controlled live verification only with explicit authorization. Until then, show support enquiries honestly and label payment integration as not live; never fabricate receipts.

Provider choice: Confirm actual organizational location and donor markets before choosing a payment service. The coding agent must use current official integration guidance rather than assume a provider account can be opened or used in every country.

Phase 11 | Administration and communication

Deliver a private staff workspace for exhibition reviews, reports, discussion interventions, mentor verification, applications, matches, campaigns, finance records, volunteer applications, contact requests, news and team editing. Show status, assignee, dates and action history. Require a reason for rejection, removal and funding changes.

Use a durable queue for notifications and email with retry limits and duplicate prevention. Implement reply, moderation, match and payment notifications; preferences govern optional messages. Marketing consent remains separate. Staff can inspect delivery failures without viewing credentials.

Completion gate: Each staff role can complete its own work and cannot perform another role's restricted actions through direct requests. Admin-only pages and documents are not included in public search, sitemaps or analytics payloads.

Phase 12 | Cross-feature quality and security checks

Area	Required evidence before release
End-to-end journeys	Registration; approved exhibition and reply; hosted text/audio conversation; mutual match; general contribution and campaign sponsorship; volunteer and mailing flows.
Access controls	Cross-account read/write attempts fail; staff-role changes are protected; private files cannot be fetched by public URL; service secrets are absent from client builds.
Failure recovery	Upload retry, dropped connection, duplicate message, expired audio token, email outage, provider webhook retry and payment reversal behave correctly.
Mobile and accessibility	Real phone checks plus keyboard and screen-reader spot checks; labelled forms, visible focus, usable menus, alt text, microphone status and text alternatives.
Performance	Test representative pages on throttled mobile connectivity. Compress media, paginate feeds, lazy-load nonessential modules and avoid autoplay. Record results and fix concrete bottlenecks.
Content and search	Correct page titles and share previews; sitemap includes only published public pages; private pages are excluded and authenticated, not merely hidden from indexing.
Operations	Backups restore into staging; monitoring and cost alerts reach the owner; logs omit secrets and sensitive documents.

Completion gate: Automated checks cover critical permissions, payment deduplication and state transitions; manual evidence covers device and audio behavior. No unresolved defect allows private-data exposure, incorrect money totals or broken primary user journeys. Record remaining minor issues explicitly.

Check the actual data collection against the privacy policy. Define retention for applications, contacts and logs, and an account deletion/request process with appropriate exceptions for financial records. Establish the audience age policy before launch rather than silently collecting sensitive data from children.

Phase 13 | Staging beta and production launch

13.1 Run an invited beta. Use 20-30 invited testers representing students, researchers, hosts, mentors and staff. Supply task-based scenarios, capture failures and fix the primary journeys. Obtain permission for any real member content that will remain at launch.

13.2 Prepare production. Configure the confirmed domain, HTTPS, production auth callbacks, service origins, email authentication, storage policies and payment/audio webhooks. Load approved content only. Disable seed/demo accounts and remove staging credentials. Document a rollback release and restore procedure.

13.3 Release review. Give the founder a concrete preview with the seven-tab navigation, passed gates, remaining limitations, operating costs and launch checklist. Obtain authorization for production changes and any real-money verification where it has not already been provided. Do not purchase services or change unrelated DNS records without authorization.

13.4 Deploy and verify. Back up before migrations; apply reviewed backward-compatible changes and deploy the tested version. Smoke-test routes, auth, uploads, audio, email and provider health. Observe logs and roll back the application if essential journeys fail. Do not blindly reverse migrations that could destroy data.

Completion gate: Production is reachable on the correct domain, core journeys work with production configuration, monitoring is active, content is approved and an operations owner can handle reports and financial exceptions.

Phase 14 | Handoff and first month of operation

Deliver: source repository and release identifier; setup README; architecture and data diagram; migration and permission notes; environment-variable names; service ownership list; deployment and rollback runbook; backup/restore procedure; test evidence; cost assumptions; staff guide; and a truthful known-issues list. Transfer credentials through the approved secret manager, never in this PDF or source control.

During the first week, review errors, unresolved reports, failed email, audio failures and payment reconciliation daily. During the first month, review usage, retention, mentor response, support workload and service spending weekly. Adjust quotas and onboarding from observed use before adding features.

Track outcomes: Member activation, approved exhibitions and repeat contributors, discussion participation, audio attendance, match acceptance/completion, reconciled contributions and project milestones. Report real values only and keep private participant data out of public dashboards.

Final instruction to Claude Code / Codex

Start with Phase 0. Work through the numbered phases and their completion gates. Build usable, connected features backed by real persistence and permissions. Keep a clear record of blocked external dependencies, complete unaffected work, and never call an unverified integration complete. Finish by providing the founder with the working release and the operational handoff above.